

LKS 2020

TEKNIK PERANCANGAN MODEL

3 DIMENSI (3D GAME ART)

Contents

Contents-----	2
Introduction to Test Project-----	3
Introduction-----	3
Description of Project and tasks-----	4
Intstructions to the Competitor-----	11

Introduction to Test Project

The following is a list of sections or information that must be included in all Test Project proposals that are submitted to LKS.

- Contents including the list of all documents, drawings, and photographs that make up the Test Project
- Introduction/overview
- Short description of project and tasks
- Instructions to the Competitor
- Equipment, machinery, installations, and materials required to complete the Test Project
- Marking scheme (incl. assessment criteria)
- Other

Introduction

The target of this competition is to develop assets for the newest version of the game, Indonesia Intelligent Agent.

Game Background.

Where the new Virus created by the terrorist, many people in Indonesia need to be a cure, the Indonesia government sends the best Intelligent Agent to find the cure, and fight the terrorist. It's not easy to defeat the terrorist because the terrorist has the best technology. But for the country, we love Indonesia Intelligent Agent must do their best for this nation.

The Brief

Through the competition, you will be creating two Assets for the game:

- Major Asset: **Indonesia Intelligent Agent (Just Choose one of them)**
 - o This character must have high-end technology
- Minor Asset: **a Weapon Companion (Just Choose one of them)**
 - o Its can be a drone
 - o Its can be Turret
 - o Its can be Mechanical Robot

Indonesia Intelligent Agent

Agent Bejo - Striker



Agent Suci - Medic



Agent Tejo – Tanker



Weapon Companion

Turret Robot



Drone



OVERVIEW WITH SCHEDULE

Result for this Competition will including the following:

Day	DELIVERABLE	DESCRIPTION
1	Modelling	- Sculpt the character (Indonesia Intelligent Agent) - o Must be high poly and view presentation must from the front, left, right, 3/4 - Rough model for props (Weapon Companion) - o The model with a maximum of 10.000 triangles
2	Modeling and UV unwrapping	- Finish Character (Indonesia Intelligent Agent) - o The model with a maximum of 50.000 triangles - Finish props (Weapon Companion) - o The model with a maximum of 10.000 triangles - Unwrapping Character - Unwrapping Props
3	Texturing and Presentation	- Texture Character - o 4096 x 4096px texture map - o Ambient occlusion, normal and opacity maps - o Diffuse/Metal/Rough PBR - Texture props - o 1024 x 1024px texture map - o Ambient occlusion, normal and opacity maps - o Diffuse/Metal/Rough PBR - Presentation - o The character must have pose with a weapon companion - o The character and companion must presentation from the front, left, right and 3/4 view - o The character and companion must presentation from the front, left, right and 3/4 view with Wireframe

NOTE – ALL ASSET ARE TO BE PLACED IN FOLDER ON THE SERVER

Description of project and task

Day 1 – Modeling

Day 1 Brief

You will be creating Indonesia Intelligent Agent, and **today you need to sculpt for Character (Indonesia Intelligent Agent) and rough 3D model for props (weapon companion) process.** Following this, you need to create a 3D sculpt and dummy model for the competition.

The committee will provide:

- The concept art for Character and props
- The base character model male and female (can create from zero)
- Presentation template

Day 1 Deliverables

By the end of today, you will need to deliver :

1. 3D Sculpt for the character such as the concept art
2. The base model for the weapon companion
3. 3D Sculpt Presentation view from the front, Side, 3/4 for the character.
4. Base model presentation view from the front, Side, 3/4 for a weapon companion.

NOTE – ALL ASSET ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 1 General Guidelines

Description

- 3D Sculpt for the character filename structure must be like **“CHR_NameCharacter_Sculpt”**
- The base model for a weapon companion structure must be like **“PRD_NameOfWeapon_Dummy”**
- the file extension must be support for Maya, 3dmax, blender, ZBrush, or related 3D software support 3D sculpt i.e **.fbx, .obj, .ztl, .zbr**

Day 2 – Modeling and UV unwrapping

Day 2 Brief

The 3D sculpt for character and base model weapon must be complete. It is not important in which order you complete the models as long as they are both ready for marking by the end of today. 3D Sculpt character should be retopology and must remain within the 50.000 triangle limit.

The base model of the weapon companion has to be complete today. Both of them must be UV unwrapped.

Day 2 Deliverables

1. Character (Indonesia Intelligent Agent) – Maximum 50.000 triangles
2. Props (A weapon companion) – Maximum 10.000 triangles
3. Both of them must be UV unwrapping

NOTE – ALL ASSET ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 2 General Guidelines

Description

1. Maya, Max or Blender Scene Organisation – only relevant assets in the scene
2. Object pivot must in zero grid coordinate (0,0,0)
3. The pivot must in below of the object
4. Unit in Scene must be 10cm/grid
5. Topology and edge loop distribution for character and props. Consider areas that require deformation
6. Both 3D models conform to edge flow that accentuates the represented asset.
7. (major/minor) Assets can be understood in silhouette only
8. No N-gons present on the assets.
9. No flipped normal present on the assets.
10. Chamfered edges on any 90 degrees on the assets
11. The model utilizes 90+% of the triangle budget on the assets.
12. Model is within triangle budget
13. Character filename structure **“CHAR_NameCharacter_MDL”**
14. Props filename structure **“PROP_NameOfWeapon_Dummy”**
15. the file extension must be support for Maya, 3dmax, blender i.e **.fbx, .obj**

Day 3 Texturing and Presentation

Day 3 Brief

Texture the character and props, character texture sheets of 4096 x 4096px (Indonesia Intelligent Agent) and 1024 x 1024px (A weapon companion). Do not make the assets share 1 sheet.

Make the most out of each sheet – note that marks are awarded for the effective use of the texture sheets.

Create texture for each asset, though you may also paint detail and create maps in 3d painting software such as Substance Painter, Photoshop, Substance Designer, or 3d Coat. Think about the variety of materials that may be used, for example, metal, plastic, skin, fiber, etc.

Make sure to follow the look and feel of the reference images.

Day 3 Deliverables

1. texture map for the character (Indonesia Intelligent Agent) and Props (weapon companion)
 - Diffuse map
 - Metal map
 - Rough map
 - Ambient Occlusion (AO) map
 - Normal map
 - Opacity map
2. 2 texture sheets
 - Character – 4096 x 4096px
 - Props – 1024 x 1024px
3. Final 3D model for the character and props with the final texture
4. Character and props must be posed for presentation
5. Character and props must be a presentation from perspective view (3/4 front and 3/4 back)

NOTE – ALL ASSET ARE TO BE PLACED IN FOLDERS ON THE SERVER

Day 3 General Guidelines

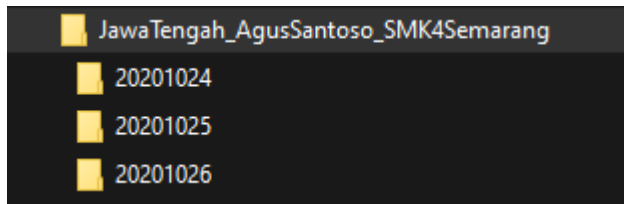
Description

1. Aspect Ratio of texture sheets are correct – **4096 x 4096(major) & 1024 x 1024(minor)**
2. PSD saved as PNG format and applied to the mode
3. UV Tidiness and Efficiency - character (Indonesia Intelligent Agent)
4. Smooth and even UV shells
5. Texel density – the size of polygons relative to the pixel resolution – Character/Pops
6. Group shells with similar colors together
7. UV coordinates exported to PS or Substance
8. UV Tidiness and Efficiency – Props (weapon companion)
9. No overlapping UVs (except for mirror/repeat)
10. UVs utilize rotation to maximize space
11. No obvious pixelation on the surface
12. No obvious stretching of textures on the surface
13. Colour and Tone represents the base color of the material
14. Surface Textures - describes materials correctly
15. The texture looks seamless on model
16. The texture is consistent with an art style
17. Specular or Metal map created
18. Normal map created
19. Roughness or smoothness map created
20. A variety of physical materials have been represented
21. Appropriate use of Opacity map created for transparency
22. Ambient Occlusion map created
23. 2 or more base colors have been used on the texture sheet
24. Height or Displacement map created
25. Smart masks used in Substance Painter
26. Evidence of variation in PBR textures (settings)
27. For character, the filename must be **“CHAR_nameCharacter_FINAL”**
28. For props, the filename must be **“PROP_nameCharacter_FINAL”**
29. Texture filename for character must be :
 - Diffuse = **“CHAR_nameCharacter_DIFF”**,
 - AO = **“CHAR_nameCharacter_AO”**,
 - Metal = **“CHAR_nameCharacter_MTL”**
 - Rough = **“CHAR_nameCharacter_RGH”**
 - Opacity = **“CHAR_nameCharacter_OPT”**
 - Normal = **“CHAR_nameCharacter_NRM”**
30. Texture filename for props must be :
 - Diffuse = **“PROP_nameCharacter_DIFF”**,
 - AO = **“PROP_nameCharacter_AO”**,
 - Metal = **“PROP_nameCharacter_MTL”**
 - Rough = **“PROP_nameCharacter_RGH”**
 - Opacity = **“PROP_nameCharacter_OPT”**
 - Normal = **“PROP_nameCharacter_NRM”**

Instructions to the Competitor

Create a folder on the server provided by the committee

The structure folder example must be like in the below:



RootFolder must be named **“yourProvince_YourName_YourSchool”**

The folder must be **“Year + Month + Day”** the example folder name must be **“20201024”**

Year = 2020

Month = 10

Day = 24

Every Final progress must upload in that folder

Internet Access

- Need internet access

Remember THE STANDARDS SPECIFICATION